

AR Wardrobe Trial

A 2D Camera-Based Virtual Dressing Room Using Pose Estimation and Person Segmentation

1. Introduction

The **AR Wardrobe Trial** is a 2D camera-based virtual dressing room that allows users to try different outfits virtually without physically wearing them. The system uses a webcam or mobile camera to capture the user and computer-vision techniques to detect their body position and identify the person within the camera frame.

The proposed system combines **MediaPipe Pose** for detecting body landmarks such as shoulders, elbows and hips with **person segmentation** for identifying the user's body region and improving the visual layering of virtual clothing.

The selected clothing items are represented as transparent 2D images and are dynamically resized, positioned and overlaid according to the user's detected body measurements and movements.

2. Problem Statement

Traditional online clothing shopping does not allow users to physically see how an outfit may look on their body before purchasing it. This can lead to uncertainty regarding appearance, size and fit, as well as increased product returns.

Physical trial rooms, on the other hand, require users to visit a store and manually change outfits.

Therefore, there is a need for an accessible virtual dressing system that allows users to try outfits through a camera, adjust clothing according to body position, receive an estimated size recommendation, evaluate apparent fit, and interact using simple gestures.

3. Proposed Solution

The proposed system will use a webcam to capture the user's live video. Each frame will be processed using two major computer-vision components.

MediaPipe Pose: detects important body landmarks such as the left and right shoulders, elbows, wrists, hips, knees and ankles. These landmarks will be used to estimate the position, orientation and approximate dimensions of the user's body.

Person Segmentation: creates a mask separating the user from the background. This information will be used to improve the placement and layering of virtual clothes so that the outfit appears more naturally integrated with the user's body.

The detected information will then be combined with transparent 2D clothing images to create the virtual try-on experience.

4. System Workflow

Camera Input → OpenCV Frame Capture → MediaPipe Pose Detection → Body Landmark Extraction → Person Segmentation → Body Measurement & Position Estimation → Select 2D Clothing Item → Resize and Position Clothing → Combine Clothing + Person Mask → Virtual Try-On Display → Fit Evaluation & Size Recommendation

5. Main Features

5.1 Real-Time Virtual Outfit Trial

Users can select clothing items such as T-shirts, shirts, tops, jackets and dresses. The selected clothing will be displayed over the user's body in real time.

5.2 Pose-Based Clothing Placement

The system will use shoulder and hip landmarks to determine where the clothing should be positioned. Shoulder distance can be used to estimate clothing width, while shoulder/hip positions can be used to determine clothing placement. As the user moves, the virtual clothing can adjust accordingly.

5.3 Person Segmentation

Segmentation will identify the user's body region separately from the background. This can improve the visual realism of the virtual outfit by allowing appropriate foreground/background layering.

5.4 Size Recommendation

The system can estimate body proportions using detected landmarks and compare them with predefined clothing-size ranges. The output can be an approximate recommendation such as **Recommended Size: M**. The recommendation will be presented as an estimate rather than an exact physical measurement.

5.5 Virtual Fit Evaluation

The system will compare estimated body dimensions with the dimensions associated with the selected clothing and provide a simple result such as **Good Fit**, **Slightly Loose** or **Slightly Tight**.

5.6 Gesture-Based Interaction

Hand or pose landmarks can be used to implement simple touchless controls. For example, a hand movement can browse outfits, a gesture can select an outfit, another gesture can change clothing, and an exit gesture can close trial mode.

6. Technologies Required

Software: Python, OpenCV, MediaPipe, NumPy, Pillow (PIL), and optionally Tkinter, Flask or Streamlit for the user interface.

Hardware: Laptop/PC with webcam or built-in camera. A mobile camera can be considered for future deployment.

7. Clothing Dataset

The system will use a collection of transparent-background 2D clothing images. Each clothing item can contain metadata such as clothing type, approximate size range and category. PNG images with transparent backgrounds are preferred for clean overlay.

8. System Architecture

USER → WEBCAM → OPEN CV → LIVE VIDEO FRAME → MediaPipe Pose + Segmentation → Body Landmarks + Person Mask → Body Position & Scale → Clothing Selection → 2D Clothing Overlay → Fit Evaluation → Size Recommendation → USER DISPLAY

9. Expected Outcome

The final system will provide a working 2D virtual dressing room where a user can stand in front of a camera and virtually try different clothing items.

The system is expected to detect the user's body in real time, track body movement, identify the person, position and scale virtual clothes, switch between outfits, provide an approximate size recommendation, provide basic fit evaluation, and support touchless interaction through gestures.

10. Unique Selling Point

The major USP is that the system is not limited to simply placing a clothing image over a person. It combines **real-time pose tracking + person segmentation + virtual outfit overlay + size recommendation + fit evaluation + gesture-based interaction** in a single lightweight **2D camera-based system**.

This makes the project practical for online shopping applications while avoiding the computational complexity of a full 3D virtual try-on system.

11. Scope

Current Scope: upper-body clothing, front-facing camera input, 2D transparent clothing assets, real-time pose tracking, person segmentation, basic size recommendation, basic fit evaluation and gesture-based outfit selection.

Future Scope: full-body clothing, pants and skirts, more accurate body measurements, improved cloth deformation, AI-based outfit recommendations, multiple body poses, mobile application deployment, personalized fashion recommendations and e-commerce integration.

12. Limitations

Since this is a **2D virtual try-on system**, it will not reproduce the complete physical behavior of real fabric. Accuracy can also depend on camera quality, lighting, user distance, body orientation, clothing-image quality and accuracy of estimated measurements.

Therefore, size and fit results should be treated as **recommendations/estimates** rather than guaranteed physical fit.

13. Conclusion

The **AR Wardrobe Trial** proposes a practical 2D virtual dressing-room solution using computer vision. MediaPipe Pose will provide body landmarks for determining the position and scale of virtual clothing, while person segmentation will provide information about the user's body region for improved visual compositing.

By combining virtual outfit trial, approximate size recommendation, fit evaluation and gesture-based interaction, the project aims to provide a more interactive and useful shopping experience than a basic clothing-overlay application.

The project demonstrates the practical application of **computer vision, image processing, augmented reality concepts and human-computer interaction** in the fashion and e-commerce domain.